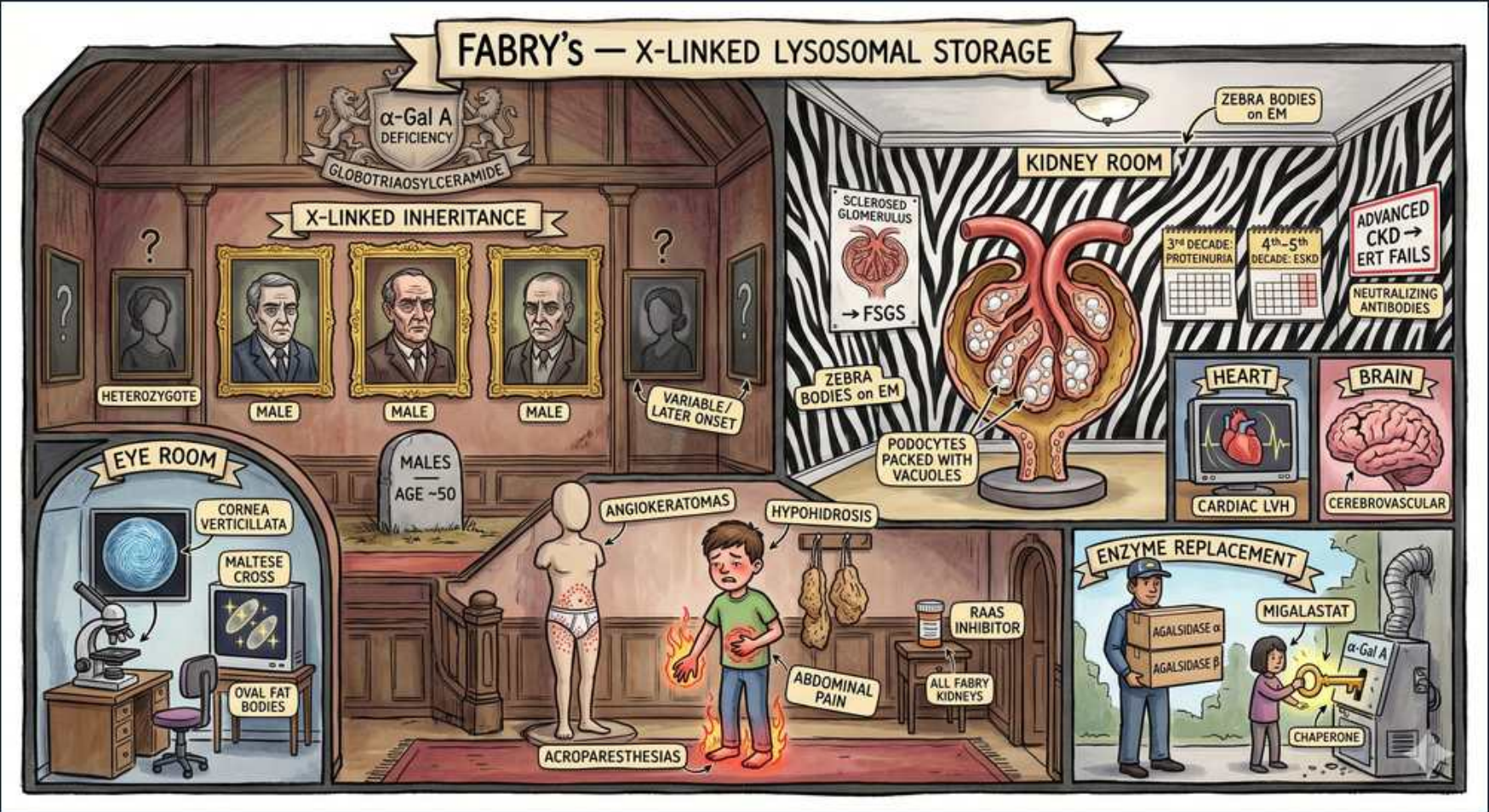


Glomerular (Nephrotic) — Fabry's Disease



1. X-linked lysosomal storage

WHAT YOU SEE

Top banner FABRY’S — X-LINKED LYSOSOMAL STORAGE

WHAT IT ENCODES

Fabry disease is an X-linked lysosomal storage disorder (GLA gene on Xq22) causing systemic glycosphingolipid accumulation. It is the second most common lysosomal storage disease after Gaucher. Because storage is in vascular endothelium, podocytes, cardiomyocytes, and neurons, it is a true multisystem disease — kidney, heart, brain, skin, eye. Mnemonic: Fabry = 'Fa-X-bry' (X-linked) versus the autosomal-recessive crowd.

BOARD TRAP

WRONG ANSWER: 'autosomal recessive inheritance like most lysosomal storage diseases.' Discriminator: Fabry is the major X-linked exception — almost all other lysosomal storage diseases (Tay-Sachs, Gaucher, Niemann-Pick, metachromatic leukodystrophy, Krabbe) are autosomal recessive. Hunter syndrome is the other classic X-linked lysosomal disease; if a boards stem says 'X-linked storage' with renal/cardiac/skin disease, pick Fabry, not Gaucher.

2. α-Gal A deficiency

WHAT YOU SEE

Top-left crest 'α-Gal A deficiency — glucotriaosylceramide'

WHAT IT ENCODES

Deficiency of the lysosomal enzyme alpha-galactosidase A (α-Gal A) blocks cleavage of the terminal galactose from globotriaosylceramide (Gb3, also called ceramide trihexoside/GL-3), so Gb3 accumulates in lysosomes throughout the body. Plasma/urine globotriaosylsphingosine (lyso-Gb3) is the key biomarker for diagnosis and monitoring. Diagnosis in males is by low leukocyte α-Gal A activity; in females (random X-inactivation can normalize enzyme levels) confirm with GLA gene sequencing.

BOARD TRAP

WRONG ANSWER: 'glucocerebrosidase (beta-glucocerebrosidase) deficiency.' Discriminator: glucocerebrosidase deficiency is Gaucher disease (accumulates glucocerebroside, gives Gaucher 'crumpled tissue paper' macrophages, hepatosplenomegaly, bone crises) — NOT Fabry. Fabry's enzyme is alpha-galactosidase A and the stored lipid is ceramide trihexoside (Gb3).

3. X-linked inheritance gallery

WHAT YOU SEE

Portrait gallery 'X-LINKED INHERITANCE' with affected males

WHAT IT ENCODES

X-linked: hemizygous males (one X) are fully affected with classic disease; heterozygous female carriers have variable, often later-onset disease due to skewed lyonization (random X-inactivation). An affected father transmits the mutant X to ALL daughters (obligate carriers) and NONE of his sons; carrier mothers transmit to 50% of each. There is no male-to-male transmission — a discriminating pedigree clue on boards.

BOARD TRAP

WRONG ANSWER: 'female carriers are always asymptomatic.' Discriminator: unlike many X-linked recessive diseases, Fabry female carriers frequently develop clinically significant disease (LVH, proteinuria, acroparesthesias, cornea verticillata) decades later because of variable X-inactivation. Modern teaching treats Fabry as X-linked with female penetrance, not strictly recessive — so a symptomatic woman does NOT exclude Fabry.

4. Males age ~50

WHAT YOU SEE

Gravestone in the center engraved 'MALES AGE ~50' — the tombstone marks that classic hemizygous males reach renal failure/death in the 4th–5th decade.

WHAT IT ENCODES

Classic hemizygous males develop progressive proteinuric CKD advancing to ESRD typically in the 4th–5th decade (~age 40–50). Renal disease evolves from microalbuminuria in adolescence to overt proteinuria and declining GFR in adulthood; renal failure plus cardiac/cerebrovascular events are the main causes of premature death. Early symptoms (acroparesthesias, angiokeratomas, hypohidrosis) begin in childhood years before the kidney fails.

BOARD TRAP

WRONG ANSWER: 'renal failure presents in infancy/early childhood.' Discriminator: in Fabry, end-organ failure is a LATE adult complication — childhood gives pain crises, sweating abnormality, and skin lesions, while ESRD and cardiomyopathy emerge in mid-adulthood. A neonate/toddler with ESRD points to congenital/genetic syndromes (e.g., congenital nephrotic syndrome NPHS1), not Fabry.

5. Cornea verticillata

WHAT YOU SEE

Eye room 'cornea verticillata'

WHAT IT ENCODES

Cornea verticillata = whorl-like, cream/golden corneal epithelial deposits radiating from a central point, seen on slit-lamp; it is highly characteristic and present in most affected males and many carrier females. It is essentially asymptomatic and does not reduce visual acuity, making it a useful silent diagnostic clue (also screen for posterior 'Fabry cataract' and retinal vessel tortuosity).

BOARD TRAP

WRONG ANSWER: 'the corneal whorls cause progressive vision loss / are an indication for corneal transplant.' Discriminator: cornea verticillata is painless and visually insignificant — it's a diagnostic marker, not a treatment target. Also note amiodarone and chloroquine/hydroxychloroquine produce an identical drug-induced corneal verticillata, so confirm with enzyme/genetics before attributing it to Fabry.

6. Maltese cross / oval fat bodies

WHAT YOU SEE

Microscope 'maltese cross / oval fat bodies'

WHAT IT ENCODES

Urinalysis shows lipid-laden tubular/epithelial cells (oval fat bodies) and free fat that give a 'Maltese cross' birefringence under polarized light. In Fabry the stored material is globotriaosylceramide; in generic nephrotic syndrome the Maltese crosses are cholesterol/lipiduria from heavy proteinuria. Look also for mulberry cells/bodies (lipid-laden cells) which are more specific to Fabry.

BOARD TRAP

WRONG ANSWER: 'Maltese crosses confirm nephrotic-range cholesterol loss and are specific for membranous nephropathy.' Discriminator: Maltese-cross birefringent lipiduria is NON-specific — it occurs in any heavy nephrotic state. It does not by itself identify the cause; the Fabry diagnosis rests on low α-Gal A activity, elevated lyso-Gb3, zebra bodies on EM, and GLA mutation, not on the urinary fat alone.

7. Angiokeratomas

WHAT YOU SEE

Skin lesions labeled ANGIOKERATOMAS over groin/trunk

WHAT IT ENCODES

Angiokeratoma corporis diffusum = clusters of small, dark red-to-purple, non-blanching papules concentrated in the 'bathing-suit' distribution (lower trunk, groin, buttocks, umbilicus, thighs). They are dilated superficial dermal capillaries with overlying hyperkeratosis and increase in number/size with age. Combined with acroparesthesias and hypohidrosis in a young male, they are the classic triad pointing to Fabry.

BOARD TRAP

WRONG ANSWER: 'these papules are Campbell de Morgan cherry angiomas / petechiae of a bleeding disorder.' Discriminator: cherry angiomas are scattered and age-related with no systemic disease; Fabry angiokeratomas cluster in the bathing-suit zone and travel WITH neuropathic pain, hypohidrosis, and proteinuria. Their distribution + accompanying multisystem clues is what flags Fabry rather than a benign vascular lesion.

8. Acroparesthesias

WHAT YOU SEE

Burning hands/feet figure 'ACROPARESTHESIAS' with flames

WHAT IT ENCODES

Acroparesthesias = episodic burning/lancinating neuropathic pain in the hands and feet, often the EARLIEST symptom (childhood/adolescence), triggered by heat, fever, exercise, or stress ('Fabry crises'). It reflects small-fiber peripheral neuropathy from Gb3 in dorsal root ganglia. Managed with neuropathic agents (carbamazepine, gabapentin, pregabalin) plus avoidance of triggers — symptomatic relief alongside enzyme replacement.

BOARD TRAP

WRONG ANSWER: 'inflammatory polyarthritis / large-joint arthralgia (think rheumatologic disease).' Discriminator: Fabry pain is NEUROPATHIC and acral (burning in fingers/toes), not synovitis — no joint swelling, normal inflammatory markers. A febrile child with burning extremity pain misdiagnosed as 'growing pains' or rheumatic fever is a classic missed-Fabry stem.

9. Hypohidrosis

WHAT YOU SEE

Tag 'HYPOHIDROSIS' pinned on the boy standing in flames — placed beside the burning figure to show reduced sweating drives the heat intolerance and heat-triggered pain crises.

WHAT IT ENCODES

Hypohidrosis/anhidrosis (reduced sweating) results from autonomic dysfunction and Gb3 deposition in sweat glands, causing heat and exercise intolerance and contributing to the heat-triggered pain crises. It is part of the early symptom complex and helps separate Fabry from other causes of acral pain.

BOARD TRAP

WRONG ANSWER: 'hyperhidrosis (excessive sweating) is the autonomic feature.' Discriminator: Fabry causes DECREASED sweating (hypohidrosis), not increased. This is why patients overheat and have exertional crises. Hyperhidrosis would point to pheochromocytoma, hyperthyroidism, or anxiety — not Fabry.

10. Podocytes vacuoles / FSGS

WHAT YOU SEE

Kidney room glomerulus 'podocytes packed with vacuoles' + 'sclerosed glomeruli → FSGS'

WHAT IT ENCODES

Gb3 accumulates in podocytes (and mesangial/endothelial cells), producing markedly enlarged, foamy, vacuolated podocytes on light microscopy; this drives progressive proteinuria and a secondary focal segmental glomerulosclerosis (FSGS) pattern leading to glomerular sclerosis and CKD. So Fabry is a storage disorder that clinically/histologically masquerades as FSGS — a key reason to send α-Gal A/genetics in unexplained proteinuric FSGS, especially young males.

BOARD TRAP

WRONG ANSWER: 'the foamy podocyte changes are idiopathic primary FSGS, so treat with high-dose corticosteroids.' Discriminator: in Fabry the podocyte vacuolation is from lipid storage (zebra bodies on EM), NOT immune/permeability-factor injury — steroids do not help and the correct therapy is enzyme replacement plus RAAS blockade. Missing this means treating a storage disease as primary glomerulonephritis.

11. Zebra bodies on EM

WHAT YOU SEE

Right note 'zebra bodies on EM'

WHAT IT ENCODES

Electron microscopy shows concentric lamellated osmiophilic inclusions — 'zebra bodies' / myelin figures — representing intralysosomal Gb3, found in podocytes, endothelium, and other cells. They are the morphologic hallmark of Fabry on renal/skin biopsy and confirm lysosomal lipid storage.

BOARD TRAP

WRONG ANSWER: 'subepithelial electron-dense deposits with spike-and-dome appearance.' Discriminator: spike-and-dome subepithelial deposits are membranous nephropathy; Fabry's EM signature is intracellular lamellated zebra/myelin bodies (storage material), not immune-complex deposits. Drug-induced phospholipidosis (amiodarone, hydroxychloroquine, aminoglycosides) can also produce myelin figures — correlate clinically.

12. Cardiac LVH

WHAT YOU SEE

Heart-monitor panel framed on the wall showing a thickened heart, labelled 'CARDIAC LVH' — the bulked-up heart on the screen depicts Gb3-laden myocardium causing concentric left-ventricular hypertrophy.

WHAT IT ENCODES

Cardiac Fabry: Gb3 in cardiomyocytes causes concentric left ventricular hypertrophy, conduction disease/arrhythmias, and a restrictive/hypertrophic cardiomyopathy phenotype; it is a leading cause of death. A characteristic clue is unexplained LVH with a relatively short PR interval and (sometimes) low-normal voltage. Fabry is an important treatable cause to screen for in 'unexplained' LVH/HCM cohorts.

BOARD TRAP

WRONG ANSWER: 'unexplained concentric LVH in a young adult = sarcomeric hypertrophic cardiomyopathy, no further metabolic workup.' Discriminator: Fabry is a phenocopy of HCM — when LVH is accompanied by proteinuria/CKD, neuropathic pain, angiokeratomas, cornea verticillata, or a positive family history with renal disease, screen α-Gal A/lyso-Gb3 because Fabry cardiomyopathy is treatable with enzyme replacement.

13. Cerebrovascular / TIA-stroke

WHAT YOU SEE

Brain panel framed beside the heart, labelled 'CEREBROVASCULAR' — the isolated brain on display flags early TIAs/strokes from Gb3 in cerebral vessels.

WHAT IT ENCODES

Cerebrovascular Fabry: Gb3 in cerebral vessels causes early TIAs and ischemic strokes (often posterior circulation) at a young age, plus white-matter lesions; vertebrobasilar dolichoectasia is a described clue. Fabry should be on the differential for cryptogenic stroke in young patients.

BOARD TRAP

WRONG ANSWER: 'a young stroke without standard vascular risk factors must be a hypercoagulable state or PFO; metabolic causes need not be screened.' Discriminator: Fabry classically causes young-onset stroke WITHOUT typical atherosclerotic risk factors — combined with proteinuria, LVH, or acroparesthesias it warrants α -Gal A testing. Failing to consider it misses a treatable hereditary cause.

14. Enzyme replacement / agalsidase

WHAT YOU SEE

Bottom-right 'ENZYME REPLACEMENT' — agalsidase α/β ' and migalastat chaperone

WHAT IT ENCODES

Disease-specific therapy = IV recombinant enzyme replacement: agalsidase beta (1 mg/kg IV every 2 weeks) or agalsidase alfa (0.2 mg/kg IV every 2 weeks). For patients with 'amenable' (responsive) GLA missense mutations, oral migalastat (a pharmacologic chaperone, 123 mg every other day) is an alternative that stabilizes residual enzyme. ERT/chaperone plus RAAS blockade (ACEi/ARB) for proteinuria and aggressive cardiovascular/CKD management is the comprehensive plan; start early before irreversible fibrosis.

BOARD TRAP

WRONG ANSWER: 'first-line treatment is corticosteroids/immunosuppression like other nephrotic glomerular diseases.' Discriminator: Fabry is a metabolic enzyme deficiency, so the specific therapy is enzyme replacement (agalsidase) or the chaperone migalastat — NOT steroids. Also, anti-drug neutralizing antibodies (more common with agalsidase beta) can blunt ERT efficacy and are a recognized limitation, but the answer is still enzyme replacement, not immunosuppression.